

SVI and SVI-JW

Stochastic-volatility-inspired slices, jump-wing handles, and constrained calibration

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Abstract

This note documents the Vol-Fitter’s SVI slice model. Raw SVI parametrizes total implied variance as a hyperbola in log-moneyness, $w(k) = a + b\{\rho(k - m) + \sqrt{(k - m)^2 + \sigma_{\text{SVI}}^2}\}$, with five parameters that are geometrically opaque but analytically convenient. The *jump-wing* (SVI-JW) re-coordinatization replaces them with five quantities in trader vocabulary — ATM variance, a time-scaled ATM skew, two normalized wing slopes, and the minimum variance. Product status up front: JW is the analytical and benchmark coordinate system of this note; production fits and stores *raw* SVI and displays model-agnostic ATM handles. Five-handle JW entry, bumping and export are not yet wired UI features, but their required analytics spine now ships: a domain-guarded converter with structured failures (`jw_to_raw_checked`) and a desk-unit ticket layer (ATM vol, 25/10 Δ risk reversals and butterflies, wing slopes, var swap, and the forward-error bump). We give the exact conversion in both directions together with its *regular inverse domain*: on $v > 0$, $p, c > 0$, $-p/2 < \psi < c/2$, $\psi \neq 0$, $\tilde{v} < v$ the displayed formulas define the unique nonsingular inverse — while the image also contains the singular stratum $\psi = 0$ (ATM at the vertex, $\tilde{v} = v$), attained by infinitely many raw slices, because the handles omit ATM curvature and cannot identify (m, σ_{SVI}) there. The benchmark converter carries *no* guards, so the domain is the caller’s contract, stated precisely and test-locked. We restate the two *coded* no-arbitrage fences in JW units — Lee’s cap becomes $\max(p, c)\sqrt{v\tau} \leq \beta_{\text{max}}$, with the production cap strictly buffered at $\beta_{\text{max}} = 1.95$ because the boundary $\beta = 2$ itself admits negative tail density — and are precise about what they guarantee: finite-weight, adjustable penalties that do *not* certify Durrleman $g \geq 0$ (a classical slice passing both while arbitrageable is test-locked). What *does* certify is the belly butterfly certificate: a dense-grid g check over the traded range that every displayed slice must pass to be publishable, with a one-shot repair refit when it fails. The default optimization chart is now *structural* — $(\beta_L, \beta_R, k^*, w^*, \kappa^*)$ under lifts, so every finite iterate is strictly floor-positive and strictly Lee-clean and the penalty rows are inert — adopted over the historical raw chart by a two-round pre-registered benchmark. Both charts carry closed-form residual Jacobians; a fresh run recovers a benchmark slice through a production JW $\rightarrow$ raw $\rightarrow$ fit $\rightarrow$ JW round trip to a fraction of a vol bp, and runs $2.85\times$ faster than the finite-difference Jacobian on that microbenchmark.

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1 Product status, conventions, and why two parametrizations

Conventions. Throughout, $k = \log(K/F_T)$ is log-moneyness and $w(k) = \sigma_{\text{imp}}^2(k) \tau_T$ total implied variance, where T labels the expiry, t_T is its calendar year fraction and τ_T is the *active variance time* — calendar time dilated by scheduled events (Notes 00, 11), with $\tau_T = t_T$ when the event clock is off. Production passes τ_T (`prepared.tau`) to the SVI calibrator, not the plain calendar maturity; classical formulas below write a bare t , and wherever it plays a time role, read τ_T . The raw SVI width parameter is written σ_{SVI} so it cannot be confused with an implied volatility. A *volatility basis point* (vol bp) is 10^{-4} in volatility. LM is the Levenberg–Marquardt least-squares algorithm and `trf` scipy’s trust-region-reflective alternative; $P = 5$ is the SVI parameter count wherever per-parameter costs appear.

Product status. JW is the trader-oriented *analytical* coordinate system used by this note and by the benchmark helper (`jw_to_raw`, unguarded, whose only repository caller is a benchmark test). Current production calibration fits and stores *raw* SVI as the selected overlay family and displays three model-agnostic numeric handles — ATM vol, skew and curvature, by finite differences of the displayed curve — for every model. There is still no five-handle JW API/UI entry, no JW bumping and no JW export in the interface; but the committee revision shipped the analytics layer any such workflow is required to sit behind: the domain-guarded converter `jw_to_raw_checked` (section 3.3) and the desk-unit ticket of section 3.4 (`models/svi_jw/desk.py`). Keep this in mind whenever the prose below says what a desk “quotes” or “reads”: it describes the coordinate system and its shipped analytics, not a wired UI workflow.

SVI is the workhorse of single-slice equity-index fitting because the hyperbola

$$w(k) = a + b \left\{ \rho(k - m) + \sqrt{(k - m)^2 + \sigma_{\text{SVI}}^2} \right\} \quad (1)$$

is, in five parameters, simultaneously (i) linear in its wings — $w(k) \sim b(1 \pm \rho) |k|$ — so its wing *form* is Lee-compatible whenever the slope cap of section 4 holds, (ii) smooth and convex enough to be butterfly-free over a wide admissible cone, and (iii) cheap to evaluate and differentiate. The cost is that $(a, b, \rho, m, \sigma_{\text{SVI}})$ are geometrically entangled: no single one is “the skew” or “the ATM level”. The Vol-Fitter therefore carries SVI in *two* coordinate systems — raw for the optimizer, jump-wing for the analysis — and converts exactly between them on the domain of section 3.3.

Four tiers of “valid”. The words matter in this note, so we fix them once:

1. **Structurally valid hyperbola:** $b > 0$, $|\rho| < 1$, $\sigma_{\text{SVI}} > 0$. Guaranteed for *every* optimizer iterate by the reparametrization of section 5 — but note a stays free, so an iterate can still have negative total variance somewhere.
2. **Minimum/Lee clean:** $\min_k w(k) \geq 0$ and $b(1 + |\rho|) \leq 2$. These are the two *coded* penalties — finite-weight, configurable, zero on a clean slice.
3. **Butterfly clean:** $w > 0$ and Durrleman $g(k) \geq 0$ everywhere (with the appropriate boundary behaviour). *Not* implied by tier 2: the counterexample of section 4 passes both screens with $g < 0$.
4. **Calendar clean:** the cross-expiry ordering $w_{T_1}(k) \leq w_{T_2}(k)$ holds; supplied by the model-agnostic hinge of Note 10, again as a finite-weight penalty.

Raw SVI slices can remain arbitrageable without conditions stronger than tier 2 — this is standard in the SVI literature (Gatheral–Jacquier; the full butterfly-domain characterization is Martini–Mingone). Production therefore adds an *acceptance rule* sitting between tiers 2 and 3: the dense-grid belly butterfly *certificate* of section 4.2, which every displayed slice must pass to count as publish-ready. It certifies tier 3 *on the traded range* — the region the wing screens do not reach and the publish-time projection never touches — not on the whole line.

Invariant

1. The optimizer can only propose structurally valid hyperbolas (tier 1): $b \geq 0$, $|\rho| < 1$, $\sigma_{\text{SVI}} > 0$ hold by reparametrization, not by rejection — and under the default *structural* chart (section 5.2) every finite iterate is additionally strictly floor-positive and strictly Lee-clean, so tier 2 holds by construction and the fences are inert.
2. The two coded no-arbitrage fences (tier 2) — non-negative minimum variance and the Lee wing cap, the latter strictly buffered *below* Lee’s bound — are soft hinges that are *exactly zero* on any tier-2-clean slice (byte-identical fits). They bound, but do not certify, tier-3 butterfly cleanliness; the certificate of section 4.2 is the acceptance rule, and an uncertified displayed slice cannot become a mark.
3. The raw $\leftrightarrow$ JW conversion is exact on its *regular* domain (proposition 1); the converter itself carries no guards, so the domain is the caller’s contract (test-locked).
4. SVI is a display overlay like any other: the band, calendar, var-swap, prior and extrapolation-fence blocks share the same product targets as the other models, expressed through *model-specific* residuals (SVI works in IV units; LQD in vega-normalized price; the var-swap routes differ). Each is absent-is-byte-identical.

Heuristic

Read (1) as a hyperbola: b is the overall steepness (the asymptotic opening angle), $\rho \in (-1, 1)$ tilts it (left wing steeper than right for the equity put skew $\rho < 0$), m shifts its vertex in k , σ_{SVI} rounds off the vertex (larger σ_{SVI} = flatter bottom), and a raises the whole curve. The wings are straight lines of slope $b(1 - \rho)$ on the left and $b(1 + \rho)$ on the right — which is exactly what Lee’s bound constrains.

2 Raw SVI geometry

The derivatives used everywhere below are

$$w'(k) = b\left(\rho + \frac{k - m}{r}\right), \quad w''(k) = \frac{b\sigma_{\text{SVI}}^2}{r^3}, \quad r = \sqrt{(k - m)^2 + \sigma_{\text{SVI}}^2}. \quad (2)$$

Note $w'' > 0$ always (the slice is one globally convex curve in k , with a single minimum — the geometric fact behind the model’s rigidity on event shapes, section 7), and the wing limits are

$$\lim_{k \rightarrow -\infty} \frac{w(k)}{|k|} = b(1 - \rho), \quad \lim_{k \rightarrow +\infty} \frac{w(k)}{k} = b(1 + \rho). \quad (3)$$

The vertex (minimum total variance) sits at $k^* = m - \sigma_{\text{SVI}}\rho/\sqrt{1-\rho^2}$ with value $w(k^*) = a + b\sigma_{\text{SVI}}\sqrt{1-\rho^2}$; this minimum is the quantity the jump-wing parametrization calls $\tilde{v}t$.

3 The jump-wing handles

The SVI-JW parametrization $(v, \psi, p, c, \tilde{v})$ at expiry t is defined by ATM and wing functionals of the slice:

$$\begin{aligned}
v &= \frac{w(0)}{t} && \text{ATM variance,} \\
\psi &= \frac{b}{2\sqrt{w_0}} \left(\rho - \frac{m}{\sqrt{m^2 + \sigma_{\text{SVI}}^2}} \right) && \text{ATM skew (time-scaled),} \\
p &= \frac{b(1-\rho)}{\sqrt{w_0}}, \quad c = \frac{b(1+\rho)}{\sqrt{w_0}} && \text{put/call wing slopes (normalized),} \\
\tilde{v} &= \frac{a + b\sigma_{\text{SVI}}\sqrt{1-\rho^2}}{t} && \text{minimum variance,}
\end{aligned} \tag{4}$$

with $w_0 = w(0) = vt$.

3.1 What the handles are, in displayed units

None of the five handles is a directly plotted quantity, and misreading their units is the most common JW error. At the production variance time τ :

$$\sigma_{\text{ATM}} = \sqrt{v}, \quad \partial_k \sigma_{\text{imp}}(0) = \frac{\psi}{\sqrt{\tau}}, \quad \beta_L = p\sqrt{v\tau}, \quad \beta_R = c\sqrt{v\tau}, \quad \sigma_{\text{min}} = \sqrt{\tilde{v}}, \tag{5}$$

where $\beta_{L,R}$ are the asymptotic *total-variance* slopes $b(1 \mp \rho)$ of (3). In words: v and $\tilde{v}$ are *variances* (the IV chart shows their square roots); ψ is a *time-scaled skew handle*, not the displayed IV tangent — the actual ATM tangent slope is $\psi/\sqrt{\tau}$; and p, c are *normalized total-variance wing slopes*, not slopes of the IV curve. Figure 1 annotates a fitted slice in exactly these displayed units, with the wing slopes on a total-variance inset where they actually live.

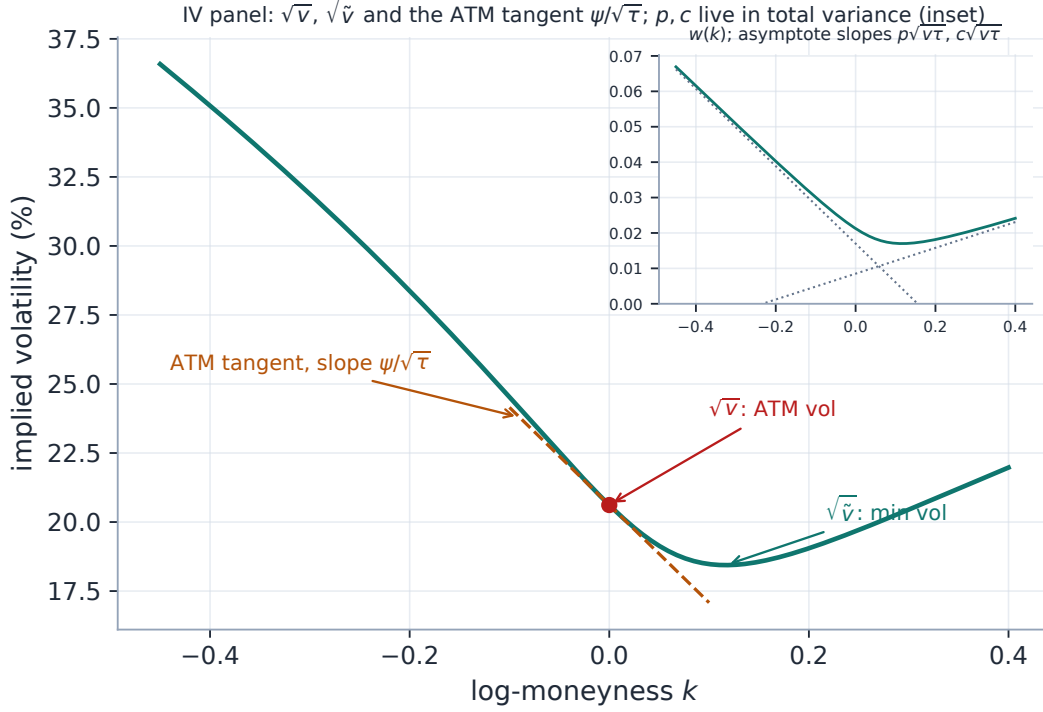


Figure 1: The five SVI-JW handles in *displayed* units. Main panel (IV): the ATM level $\sqrt{v}$, the minimum $\sqrt{\tilde{v}}$, and the ATM tangent of slope $\psi/\sqrt{\tau}$. Inset (total variance $w(k)$): the two asymptotes whose slopes are $\beta_L = p\sqrt{v\tau}$ and $\beta_R = c\sqrt{v\tau}$ — the wing handles are total-variance objects, not IV slopes.

3.2 The conversion, both ways

The forward direction, jump-wing *to* raw, is what the shipped converter `jw_to_raw` computes (today its only repository caller is the benchmark test; see the product-status paragraph of section 1). Set $w_0 = vt$, $\sqrt{w_0} = \sqrt{vt}$, and

$$b = \frac{\sqrt{w_0}}{2}(p + c), \quad \rho = \frac{c - p}{c + p}, \quad \chi := \frac{m}{\sqrt{m^2 + \sigma_{\text{SVI}}^2}} = \rho - \frac{4\psi}{p + c}. \quad (6)$$

From χ one recovers $m = \chi\sigma_{\text{SVI}}/\sqrt{1 - \chi^2}$ and $\sqrt{m^2 + \sigma_{\text{SVI}}^2} = \sigma_{\text{SVI}}/\sqrt{1 - \chi^2}$; subtracting the minimum-variance identity from the ATM identity gives the closed form

$$\sigma_{\text{SVI}} = \frac{w_0 - \tilde{v}t}{b((1 - \rho\chi)/\sqrt{1 - \chi^2} - \sqrt{1 - \rho^2})}, \quad a = \tilde{v}t - b\sigma_{\text{SVI}}\sqrt{1 - \rho^2}. \quad (7)$$

The reverse map *from* raw *to* jump-wing is just the definitions (4) evaluated on the fitted raw slice; the backend ships no `raw_to_jw` — that direction lives in the figure generator and in these equations. Section 6 runs the production round trip numerically. The forward conversion (6)–(7) is direct:

Listing 1: SVI-JW handles $\rightarrow$ raw SVI (distilled from `models/svi_jw/svi.py`; agrees with it to 10^{-12}).

```

1 def jw_to_raw(t, v, psi, p, c, vt):
2     # (v, psi, p, c, vtilde) -> (a, b, rho, m, sigma)
3     w0 = v*t; sq = np.sqrt(w0)
4     b = 0.5*sq*(p + c)
5     rho = (c - p)/(c + p)
6     chi = rho - 4*psi/(p + c)      # m / sqrt(m^2 + sigma^2)
7     r1x = np.sqrt(1 - chi*chi)
8     r1r = np.sqrt(1 - rho*rho)
9     sigma = (w0 - vt*t) / (b*((1 - rho*chi)/r1x - r1r))
10    m = chi*sigma/r1x
11    a = vt*t - b*sigma*r1r
12    return a, b, rho, m, sigma

```

3.3 The regular inverse domain, exactly

Not every quintuple $(v, \psi, p, c, \tilde{v})$ is a smile, and — a finer point — *regular invertibility* is not the same as *representability*. The formulas above divide by $p + c$, by $\sqrt{1 - \chi^2}$ and by the bracket in (7), and the production converter carries *no* guards — so the exact domain is worth stating and proving.

Proposition 1 (Regular inverse domain). *Fix $t > 0$. On*

$$v > 0, \quad p > 0, \quad c > 0, \quad -\frac{p}{2} < \psi < \frac{c}{2}, \quad \psi \neq 0, \quad \tilde{v} < v,$$

the map (6)–(7) defines the unique, nonsingular inverse of the JW definitions (4): it produces a structurally valid raw slice ($b > 0$, $|\rho| < 1$, $\sigma_{\text{SVI}} > 0$, m finite), and outside this set the displayed formulas produce no such slice. In words: the wings must be genuinely upward, the quoted ATM skew must lie strictly between half the wing slopes, and the ATM variance must sit strictly above the minimum.

Two complements make the full picture. First, *representability* is larger than regular invertibility: the image of the JW map also contains the singular stratum $\psi = 0$, $\tilde{v} = v$ (remark 1 below), realized by *infinitely many* raw slices — so “the formulas fail” does not mean “no slice has these handles”. Second, a financially usable smile additionally requires $\tilde{v} \geq 0$ (tier 2 of section 1); that is an economic condition, separate from the algebraic invertibility above.

Proof. $p, c > 0$ gives $b = \frac{\sqrt{w_0}}{2}(p + c) > 0$ and $\rho = (c - p)/(c + p) \in (-1, 1)$. For $\chi = \rho - 4\psi/(p + c)$, using $(\rho - 1)(p + c) = -2p$ and $(\rho + 1)(p + c) = 2c$,

$$|\chi| < 1 \iff \frac{(\rho - 1)(p + c)}{4} < \psi < \frac{(\rho + 1)(p + c)}{4} \iff -\frac{p}{2} < \psi < \frac{c}{2},$$

which keeps $m = \chi\sigma_{\text{SVI}}/\sqrt{1 - \chi^2}$ finite. For σ_{SVI} : by Cauchy–Schwarz applied to the unit vectors $(\rho, \sqrt{1 - \rho^2})$ and $(\chi, \sqrt{1 - \chi^2})$,

$$\rho\chi + \sqrt{1 - \rho^2}\sqrt{1 - \chi^2} \leq 1, \quad \text{i.e.} \quad \frac{1 - \rho\chi}{\sqrt{1 - \chi^2}} \geq \sqrt{1 - \rho^2},$$

with equality iff $\chi = \rho$, i.e. iff $\psi = 0$. So the bracket in (7) is strictly positive exactly when $\psi \neq 0$, and $\sigma_{\text{SVI}} > 0$ then requires the numerator $w_0 - \tilde{v}t > 0$, i.e. $\tilde{v} < v$. Conversely each failure breaks a

requirement. If $p + c < 0$ then $b < 0$; if $p + c = 0$ the very first formula is undefined; and if $p + c > 0$ with one of $p, c \leq 0$ then $|\rho| = |c - p|/(c + p) \geq 1$ — so $p, c > 0$ is forced. ψ outside the band sends $|\chi| \geq 1$ (m infinite); $\psi = 0$ zeroes the bracket; $\tilde{v} \geq v$ makes $\sigma_{\text{SVI}} \leq 0$. $\square$

Remark 1 (The $\psi = 0$ stratum is a genuine coordinate singularity). On a true SVI slice, $\psi = 0$ means $\chi = \rho$, which happens exactly when the ATM point *is* the vertex — and then $\tilde{v} = v$ automatically. (Call it the *ATM-at-the-vertex* stratum: $k^* = 0$ implies no symmetry unless $\rho = 0$.) On that stratum the five handles do not identify the slice: for any $\sigma_{\text{SVI}} > 0$, the raw slice with $m = \rho\sigma_{\text{SVI}}/\sqrt{1 - \rho^2}$ and $a = \tilde{v}t - b\sigma_{\text{SVI}}\sqrt{1 - \rho^2}$ reproduces the *same* $(v, 0, p, c, v)$ — the handles omit ATM curvature, and this is where it shows (test-locked: `test_svi_domain.py::test_psi_zero_stratum_not_identified`). The jump-wing chart is a fine coordinate system *away* from the ATM-at-the-vertex stratum, and ill-conditioned near it: as $\psi \rightarrow 0$, the conversion computes σ_{SVI} as a ratio of two vanishing quantities, with catastrophic cancellation before the limit.

Caution

The benchmark `jw_to_raw` implements the formulas verbatim with *zero* domain checks, and the failure modes differ by case — none is uniformly “silent”: a scalar $p + c = 0$ raises `ZeroDivisionError`; ψ outside $(-p/2, c/2)$ takes the square root of a negative and returns NaNs (with a runtime warning); $\tilde{v} \geq v$ returns $\sigma_{\text{SVI}} \leq 0$, not a slice; and a near-zero ψ silently amplifies rounding. That contract is test-locked (`test_svi_domain.py`) — but production no longer relies on the contract alone. Committee revision R5 promoted a guarded variant, `jw_to_raw_checked`: it validates the *complete* regular domain of proposition 1 and raises a structured `JWDomainError` whose machine-readable code names the failed inequality (`nonpositive_tenor` / `nonpositive_variance` / `nonpositive_wing` / `atm_slope_out_of_range` / `singular_stratum` / `floor_not_below_level`) instead of case-dependent NaNs. It also repairs the numerics near the singular stratum: the denominator of (7) vanishes *quadratically* as $\psi \rightarrow 0$, and the checked converter evaluates it in a rearranged, algebraically identical form that never subtracts two large near-equal numbers, so the inverse stays accurate right up to the explicit rejection. Anything user-facing is required to sit behind the checked converter (test-locked, `test_svi_desk_and_guards.py`); the unguarded one remains for the benchmark.

3.4 From handles to desk units

The JW handles are model coordinates in model units — v a variance, ψ a total-volatility slope, p, c normalized variance slopes. A desk quotes none of those. The desk-unit ticket (`models/svi_jw/desk.py`, committee R5) converts *any* smile — raw SVI, LQD or MCS — into the instruments actually traded: the ATM vol $\sqrt{w(0)/t}$; 25Δ and 10Δ risk reversals and butterflies, with strikes solved on the model smile via the *forward* Black delta $\Phi(d_+)$ (no rates convention — the fitter works in forward measure throughout); the actual asymptotic wing slopes (the Lee objects, not the normalized p, c indicators); and the var-swap vol by log-contract replication. The ticket also carries the committee’s “missing derivative”: `forward_bump` re-reads the same ticket after a relative forward error dF/F — a wrong forward shifts every log-moneyness by $-dF/F$ and is easily misread as skew or wing movement, and the bump row shows exactly which desk quantities a pure forward mistake moves. This is an analytics layer, not a UI: its role is to be the conversion layer that any future JW-handle workflow is *required* to sit behind, test-locked now rather than designed later.

4 Static no-arbitrage

4.1 Butterfly: Durrleman’s function

For a positive, sufficiently smooth total-variance slice w with the appropriate wing behaviour (the large- $|k|$ limits that make the density integrate to one — satisfied by any structurally valid SVI slice with $w > 0$), the slice is free of butterfly arbitrage iff the risk-neutral density it implies is non-negative, which Durrleman’s condition expresses directly on total variance:

$$g(k) = \left(1 - \frac{k w'(k)}{2w(k)}\right)^2 - \frac{w'(k)^2}{4} \left(\frac{1}{w(k)} + \frac{1}{4}\right) + \frac{w''(k)}{2} \geq 0 \quad \forall k. \quad (8)$$

For raw SVI, g is a smooth function of (1)–(2) with no closed-form sign, and where each consumer evaluates it matters: the backtest’s arbitrage metric evaluates the closed-form (w, w', w'') triple (Note 09); the application’s extrapolated-region diagnostic finite-differences the smooth *model* w (never reconstructed option prices); this note’s figures use the closed form. The *default* SVI fit itself carries no g hinge and samples no g rows — the Gatheral–Jacquier parameter-space conditions bound the wing steepness $b(1 + |\rho|)$ and the minimum total variance, and the calibrator’s core fences are the two that usually bite.

That “usually” is load-bearing, and there is a classical counterexample: the Gatheral–Jacquier slice $(a, b, \rho, m, \sigma_{\text{SVI}}) = (-0.0410, 0.1331, 0.3060, 0.3586, 0.4153)$ has positive minimum variance (0.0116) and a Lee slope far under the cap (0.174), yet $g(k)$ dips to -0.033 near $k \approx 0.88$: both coded screens pass while genuine butterfly arbitrage remains (test-locked in `test_svi_domain.py`; full anchor in the traceability table).

How g is *measured* matters as much as what it says. An earlier backtest metric evaluated g by finite differences of reconstructed option prices, and its numerical noise flagged 28.3% of provably-clean LQD slices as arbitrageable; recomputed analytically from each model’s own (w, w', w'') , the LQD rate reads 0.0%, and SVI’s flagged rate fell from 20.8% to 9.2% — the survivors being genuine: the soft hinges fence the conditions that bite, they do not certify $g \geq 0$ pointwise. The production rule that came out of that audit is that the diagnostic is always computed on model derivatives, never by differencing prices (Note 09 has the cross-model treatment).

4.2 The belly certificate and the one-shot repair

The counterexample above is why the certificate exists: the cheap screens fence the wings and the floor, and the Vogt-type dip lives exactly where they do not reach — the *belly* of the traded range. Committee revision R2 promoted the check itself to the acceptance rule. On every displayed slice, whatever its model, production evaluates Durrleman g on an 801-point grid over the traded strike range — from the model’s own derivatives, never by differencing prices — and certifies the slice iff $g \geq -10^{-4}$ at every grid point (a finite-difference-stencil allowance, orders below any real violation: the Vogt dip is -0.033 , the Lee-boundary example of section 4.4 is -0.0485). The certificate costs a few dozen microseconds, so affordability is no argument against running it on every fit. Its verdict has teeth: an uncertified slice fails publish-readiness and hard-blocks export with an HTTP 409 — *an uncertified slice cannot become a mark*. And failure is not terminal: when a first SVI fit fails the certificate, the calibrator refits *once* with a belly hinge sampled on the traded-range grid and keeps the repair only if it certifies (`bellyRepair`, on by default); clean first fits never see a second solve. The certification case `belly_certificate` locks the pipeline end to end, Vogt slice included.

4.3 The six production layers

What production actually runs is a stack of six distinct layers, in increasing distance from the core fit:

1. **Core fences** (always in the objective): the min-variance and Lee penalty rows of section 4.4 — finite-weight, configurable, zero on a clean slice, and *structurally* zero under the default structural chart (section 5.2).
2. **Belly certificate** (always on; section 4.2): the dense-grid acceptance rule on the traded range — fails readiness and hard-blocks publish on an uncertified slice.
3. **One-shot belly repair** (`bellyRepair`, default on): a single refit with the belly hinge when the certificate fails, kept only if the repair certifies.
4. **Extrapolated-region enforcement** (`extrapEnforce`, default *off*; Notes 09–10): opt-in tapered residual rows on the extrapolated strike region — a Durrleman- g hinge on the slice’s own envelope, a tapered calendar hinge against the previous displayed slice, and the wing-slope-order hinge. These rows are finite-differenced while the core SVI rows stay analytic, so the fit runs a *hybrid* Jacobian (`calib/extrap.py`; `models/svi_jw/calibrate.py`).
5. **Advisory diagnostics** (always on): the Quality tab measures and reports remaining extrapolated-region $g < 0$ per fit — measurement, not enforcement.
6. **Publish-time wing projection** (export-only; Note 09 Phase 3): a discrete projection of the exported wing samples that raises prices only, leaving the fitted core pinned — audited to introduce no calendar crossings (the manifest’s `projectionCalendarWorstBp`).

No single layer turns a fitted SVI slice into an everywhere-certified butterfly-free object; together they fence (1), certify the traded range with teeth (2–3), bound and measure beyond it (4–5), and clean at the boundary (6).

Proposition 2 (The enforced conditions, in JW units). *Under the conversion of section 3, the two coded penalties read directly on the trader handles: the minimum-variance condition $a + b\sigma_{\text{SVI}}\sqrt{1 - \rho^2} \geq 0$ is exactly $\tilde{v} \geq 0$, and Lee’s cap $b(1 + |\rho|) \leq \beta_{\text{max}}$ is exactly*

$$\max(p, c) \sqrt{vt} \leq \beta_{\text{max}} \quad (\text{Lee’s bound is 2; the production cap is 1.95, section 4.4}).$$

Proof. The first is the definition of $\tilde{v}$ in (4) times $t > 0$. For the second, $b(1 \mp \rho) = p\sqrt{w_0}$ and $c\sqrt{w_0}$ respectively by (4), so $b(1 + |\rho|) = \max(b(1 - \rho), b(1 + \rho)) = \max(p, c)\sqrt{w_0}$ with $w_0 = vt$. $\square$

So a desk can sanity-check a handle quote before any conversion: wings non-negative at the vertex ($\tilde{v} \geq 0$), skew inside $(-p/2, c/2)$ (proposition 1), and $\max(p, c)\sqrt{vt} \leq 2$ (Lee).

4.4 The two soft penalties

The optimizer adds, to the data residual, two penalty rows that are *exactly zero* on a tier-2-clean slice and grow linearly past the boundary:

$$\rho_{\text{pen}}(\theta) = W \begin{pmatrix} \max(- (a + b\sigma_{\text{SVI}}\sqrt{1 - \rho^2}), 0) \\ \max(b(1 + |\rho|) - \beta_{\text{max}}, 0) \end{pmatrix}, \quad (9)$$

with residual multiplier $W = 1000$ and Lee cap $\beta_{\max} = 1.95$. The first prevents *negative total variance at the minimum*; the second caps the asymptotic total-variance slope. Because both vanish on a clean fit, a tier-2-clean smile (such as the benchmark of section 6) is byte-identical with or without them — they are a feasibility fence, not a regularizer.

The cap’s default is deliberately *not* Lee’s bound of 2, and the reason is a committee finding worth a paragraph. A slice sitting exactly at $\beta = 2$ passes the floor and wing screens with zero penalty — the hinge is zero precisely on the boundary — yet already carries negative tail density: the committee’s counterexample reads $g(10) = -0.0485$ with minimum total variance 0.44. The boundary itself is broken, and the trap was *live*: before the buffer, a production SPY wing sat at exactly 2.0000, and the buffered refit moved the fitted IVs by 0.03 vol bp — the fence moved, the smile did not. The default cap is therefore strictly buffered at 1.95; the historical 2.0 is reachable only as explicit configuration, and the certification case `svi_lee_boundary` keeps the counterexample fenced and the tail limit strictly positive.

Caution

Read the semantics of W carefully: it multiplies the *residual* rows in (9), so the least-squares objective receives an effective W^2 coefficient on the squared violation — and the two rows do not even share a natural financial unit (one is a total variance, the other a dimensionless slope), so W is a residual multiplier, not a “weight in vol² units”. It is deliberately large enough to dominate a violated constraint yet invisible on a clean slice. It is *not* a tuning dial for fit quality: raising it cannot improve a feasible fit, and lowering it only lets the optimizer wander into the arbitrageable cone. Both fences are also *adjustable*, not unconditional: a user can set `sviPenaltyWeight` = 0 (fences off) or pick a `leeSlopeMax` at or above 2 — restoring the broken boundary — so they are configuration, not guarantees. Note 09 develops the Lee bound across all the models; here it is one of these two hinges.

5 Calibration

5.1 The raw chart: unconstrained reparametrization

Rather than run a bound-constrained optimizer, the Vol-Fitter reparametrizes so that *every* point of $\mathbb{R}^5$ maps to a *structurally valid raw hyperbola* (tier 1 of section 1):

$$a \text{ free}, \quad b = \text{softplus}(\theta_b) \geq 0, \quad \rho = \tanh(\theta_\rho) \in (-1, 1), \quad m \text{ free}, \quad \sigma_{\text{SVI}} = e^{\theta_\sigma} > 0. \quad (10)$$

This guarantees $b \geq 0$, $|\rho| < 1$, $\sigma_{\text{SVI}} > 0$ for free, so the unconstrained Levenberg–Marquardt solver never wastes effort on the box and never proposes a nonsensical hyperbola. It guarantees no more than that: a stays free, so an optimizer iterate can still carry negative total variance somewhere (the vol conversion floors w at 10^{-12} so the residual stays evaluable), and arbitrage cleanliness is the business of the fences of section 4, which remain soft. This is the historical *raw* chart (`sviChart="raw"`), kept as explicit configuration for comparability and rollback; the shipped default is the structural chart below.

5.2 The structural chart (the default)

The committee’s chart parameterizes the slice by the quantities the guarantees are *about* — the two actual asymptotic wing slopes, the vertex location, the minimum total variance, and the vertex curva-

ture — each lifted so that every finite optimizer vector is admissible (`models/svi_jw/structural.py`):

$$\beta_L = \beta_{\max} \text{logistic}(\ell), \quad \beta_R = \beta_{\max} \text{logistic}(r), \quad k^* \text{ free}, \quad w^* = \text{softplus}(h) > 0, \quad \kappa^* = e^q > 0, \quad (11)$$

with $\beta_{\max}$ the buffered Lee cap of section 4.4, so both wings are *strictly* Lee-clean at every iterate. Raw recovery is exact:

$$b = \frac{\beta_L + \beta_R}{2}, \quad \rho = \frac{\beta_R - \beta_L}{\beta_R + \beta_L}, \quad \sigma_{\text{SVI}} = \frac{b(1 - \rho^2)^{3/2}}{\kappa^*}, \quad m = k^* + \frac{\sigma_{\text{SVI}} \rho}{\sqrt{1 - \rho^2}}, \quad a = w^* - b \sigma_{\text{SVI}} \sqrt{1 - \rho^2}. \quad (12)$$

Under this chart the floor and Lee penalty rows are *structurally* zero and the trial-variance clip never fires ($w(k) \geq w^* > 0$ everywhere): tier 2 holds by construction and the fences become inert bookkeeping. It does *not* certify $g \geq 0$ — the belly certificate of section 4.2 remains the acceptance authority.

The default was flipped to structural by a two-round *pre-registered* benchmark (ratified 2026-07-26; the adjudication lives in `backtest/FINDINGS_svi_chart.md`), and its decisive finding is a caution for anyone comparing fitting charts. On $\sim 29\text{k}$ real fits per arm the raw chart’s *headline* arbitrage incidence looked lower — but a third of its fits (9,472) had exhausted the 500-evaluation budget before reaching any optimum, and those unfinished fits, whose arb rate reads 0.317% precisely *because* they are unfinished, diluted the average: a survivorship artifact in reverse. Conditioned on convergence, raw reads 1.076% genuine-arb incidence against structural’s 0.864% (all structural fits converge; 0.822% on the matched converged population), with structural better or equal in all 12 regime-precision medians, zero structural breaks, 594 vs 9,472 evaluation-cap exhaustions, and $\sim 3\times$ faster wall-clock. The offline backtest’s frozen SVI-JW arm stays raw at cap 2.0 forever, for comparability of its historical parts.

One numerical honesty note from the adoption follow-up: lifting through logistics puts the chart’s edges at floating-point saturation, and two latent boundary bugs surfaced only when the analytic Jacobian landed. A wing saturated against an $O(1)$ other can round ρ to an exact ± 1 — collapsing σ_{SVI} to 0 and m to NaN — and $\text{logistic}(x)$ rounds to exactly 1.0 for $x \gtrsim 37$, landing a wing exactly *at* the cap and breaking the strict-fence contract. Both are closed structurally: the lifts are clipped one ulp inside 1, and $1 - \rho^2$ is computed by the exact identity $4\beta_L\beta_R/(\beta_L + \beta_R)^2$, which never rounds to zero.

5.3 Objective, weights and the data-driven start

The data term is a plain implied-volatility residual, $r_i = \sqrt{\omega_i}(\sigma^{\text{SVI}}(k_i) - \sigma_i)$. The surfaced weighting choices are `weightScheme` $\in \{\text{equal}, \text{tv_density}\}$ (Note 07) — the same two schemes every model gets; the calibrator’s low-level API also accepts arbitrary ω_i (e.g. vega^2 , used by tests and research scripts), but that is a code capability, not a surfaced setting. The initializer is geometric and reads the smile directly: the argmin of w seeds (a, m) , and the two finite wing slopes seed (b, ρ) via (3), then the active chart is inverted to get θ_0 (the structural chart re-expresses the *same* data-driven start). On a liquid smile a single LM pass converges from this start.

One adversarial guard sits before any solve: with fewer than three usable quotes the calibrator *refuses deterministically with a reason* — five parameters against fewer residual rows would otherwise crash inside scipy with an opaque message — and the caller’s fallback policy (keep the last good surface, or another family) takes over. The full adversarial battery — one-sided chains, missing ATM,

0DTE-scale variance, $|\rho| \rightarrow 1$, duplicate strikes, 50% quote noise, crossed bands, evaluation-cap exhaustion — fits finite and fence-respecting or refuses with a reason on *both* charts; it is certification case `svi_adversarial_inputs`.

5.4 The analytic Jacobian, derived

The analytic Jacobian is worth deriving rather than asserting; it is three chain-rule layers stacked (`models/svi_jw/jacobian.py`). With $x = k - m$ and $r = \sqrt{x^2 + \sigma_{\text{SVI}}^2}$, the raw-parameter partials of the hyperbola are

$$\frac{\partial w}{\partial a} = 1, \quad \frac{\partial w}{\partial b} = \rho x + r, \quad \frac{\partial w}{\partial \rho} = b x, \quad \frac{\partial w}{\partial m} = -b \left(\rho + \frac{x}{r} \right), \quad \frac{\partial w}{\partial \sigma_{\text{SVI}}} = \frac{b \sigma_{\text{SVI}}}{r}. \quad (13)$$

The data rows live in IV units, so the second layer is the Black chain rule for $\sigma_{\text{imp}} = \sqrt{w/\tau}$:

$$\frac{\partial \sigma_{\text{imp}}}{\partial w} = \frac{1}{2\sqrt{w\tau}} = \frac{1}{2\tau\sigma_{\text{imp}}}, \quad (14)$$

with the gradient zeroed wherever the $w \leq 10^{-12}$ floor is active (the clamp flattens it). The third layer is the reparametrization (10):

$$\frac{db}{d\theta_b} = \text{sigmoid}(\theta_b) = 1 - e^{-b}, \quad \frac{d\rho}{d\theta_\rho} = 1 - \rho^2, \quad \frac{d\sigma_{\text{SVI}}}{d\theta_\sigma} = \sigma_{\text{SVI}}. \quad (15)$$

The hinge rows use the subgradient convention “active linear part, else zero”: a penalty/band/calendar $\max(\cdot, 0)$ row contributes its full linear derivative where the violation is strictly positive and a zero row otherwise (at the kink itself the row is treated as inactive). Concretely, the min-variance row differentiates $-(a + b\sigma_{\text{SVI}}\sqrt{1 - \rho^2})$ — e.g. its θ_ρ entry is $+Wb\sigma_{\text{SVI}}\rho\sqrt{1 - \rho^2}$ after (15) — and the Lee row differentiates $b(1 + |\rho|)$ with the $\text{sign}(\rho)$ subgradient for $|\rho|$.

The structural chart runs the *same* rows through one more layer: the residual Jacobian is assembled in raw-parameter space and pushed through the closed-form 5×5 chain matrix $\partial(a, b, \rho, m, \sigma_{\text{SVI}})/\partial(\ell, r, k^*, h, q)$ (`structural_chain`), which mirrors the lift exactly, including the saturation-clip subgradients (a clipped column is zero). Measured on the adoption A/B against the finite-difference structural path: identical smiles (ATM equal to 10^{-6} , same maximum IV error) at $2.1\text{--}2.4\times$ wall-clock, with $\text{nfev} \leq 26$ against the FD range of 30–86. Both charts share the same gate: the calibrator uses the analytic Jacobian whenever no var-swap or prior block is present; those blocks are not differentiated and switch the fit to scipy’s finite differences (LQD, by contrast, now differentiates its var-swap rows analytically — Note 01), and the belly-repair hinge also takes the FD path (repair refits only). With `extrapEnforce` on, the fit is *hybrid*: the rows above stay analytic and only the small extrapolation block is finite-differenced.

5.5 The optional blocks

The optional blocks share the same product targets as the rest of the series, expressed through model-specific residuals:

- **Band fit** (Note 07): the mid residual becomes a vol-space band hinge with a small mid anchor.

- **Calendar hinge** (Note 10): the *model-agnostic* constraint $\sqrt{W_{\text{cal}}} \max(\text{floor} - w(k), 0)$ penalizes calendar crossings against the nearer expiry — the same product target LQD expresses via its quantile $A(z)$, here written directly on $w(k)$. Being a finite-weight hinge, it drives crossings down rather than provably to zero; the locking tests require a substantial reduction of the violation, not exact elimination.
- **Extrapolation fences** (`extrapEnforce`, Notes 09–10): the tapered extrapolated-region rows of section 4.3, default off, hybrid-Jacobian when on.
- **Var-swap target** (Note 08) and **prior persistence** (Note 13): the same targets as LQD through SVI-native residuals (the var-swap routes differ per model: LQD uses its closed-form quantile moment, SVI the strike replication) — so the SVI *display overlay* receives the same prior treatment as the primary model, closing a long-standing asymmetry in which SVI overlays got no prior at all.

Performance

The solver is Levenberg–Marquardt (`method="lm"`), not trust-region reflective. On noisy real chains LM crosses the penalty kinks in far fewer iterations than `trf` at matched tolerance, so it is the faster same-accuracy choice; the analytic Jacobian is then a drop-in that swaps `scipy`’s $1 + P$ finite-difference evaluations ($P = 5$) for one closed-form call. It shipped after the offline backtest (project `backtest/`) identified those finite-difference evaluations as SVI’s dominant fit cost. Two measurements, labeled for what they are: (i) a *historical real-node* result — $26.3 \rightarrow 10.2$ ms per fit ($\approx 2.6\times$) on spike-regime backtest nodes, June-2026 harness configuration, same optimizer and tolerances, same solution within fit precision; and (ii) this note’s *private residual/Jacobian microbenchmark* — $2.85\times$ on the 25-quote synthetic benchmark (section 6), single-threaded on the development machine, core rows only, costs agreeing to $2.7e - 33$. See appendix B.

6 Worked example: noise-free raw-SVI recovery and JW readout

The cleanest test of an SVI-JW implementation is a round trip through the shipped converter, and since this pass the figure generator runs exactly that: the target raw slice is built by the *production* `jw_to_raw` from the jump-wing handles $(v, \psi, p, c, \tilde{v}) = (0.0425, -0.25, 0.75, 0.25, 0.034)$ at $t = 0.5$, giving $(a, b, \rho, m, \sigma_{\text{SVI}}) \approx (0.010625, 0.07289, -0.5, 0.05831, 0.10100)$. Generate noise-free quotes from it, refit raw SVI from the geometric start, and read the handles back through the JW definitions — a JW→raw→fit→JW loop in which the forward leg is production code.

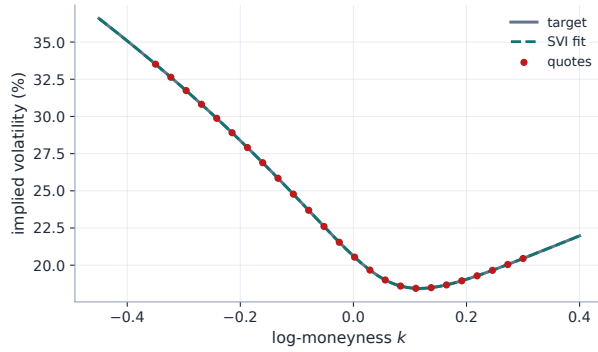
The fit recovers the raw parameters (table 1) with a maximum implied-volatility error of 0.00 volatility basis points in 21 residual evaluations, the wing slopes come out $b(1 - \rho) = 0.1093$ (put) and $b(1 + \rho) = 0.0364$ (call), and — the point of the exercise — the recovered jump-wing handles (table 2) reproduce the original $(0.0425, -0.25, 0.75, 0.25, 0.034)$ to display precision. The conversion (4)–(7) is therefore exact on this benchmark, not approximate. (The generator runs the raw chart; on clean quotes like these the structural chart reaches the same optimum to solver tolerance — chart choice moves the trust-region geometry, not the target — which is test-locked.) Figure 2(a) shows the fit and fig. 2(b) the Durrleman function staying non-negative on the diagnostic grid $k \in [-0.45, 0.40]$.

Table 1: Recovered raw SVI.

Parameter	Value
a	+0.010625
b	+0.072887
ρ	-0.500000
m	+0.058310
σ	+0.100995

Table 2: Recovered SVI-JW handles.

Handle	Value
v	+0.042500
ψ	-0.250000
p	+0.750000
c	+0.250000
$\tilde{v}$	+0.034000



(a) target, fit, quotes

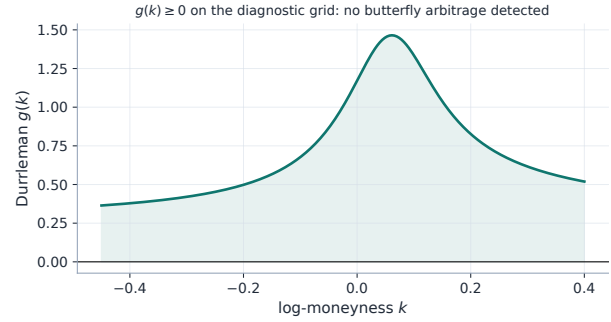
(b) Durrleman $g(k) \geq 0$

Figure 2: Left: the benchmark smile and its SVI fit (max error 0.00 vol bps). Right: the butterfly diagnostic $g(k)$, non-negative on the plotted diagnostic grid.

Heuristic

A recovery error of 0.00 vol bps is expected because the target *is* a noise-free SVI slice — the exercise validates the optimizer and the conversion, not the expressiveness of SVI, and says nothing about market fit. On real NBBO history the published spike-regime replay measured SVI-JW at 24.3 vol bp RMS in-sample and 26.8 out-of-sample across 1,576 nodes (`backtest/results`); SVI is also stiffer than LQD or Local-Vol on event or double-hump shapes (one globally convex hyperbola, one vertex), which is precisely why the app offers the other models.

7 What is original here, and limitations

SVI and the jump-wing handles are due to Gatheral; the arbitrage-free analysis is Gatheral–Jacquier and Durrleman. The implementation choices worth noting are the *structural chart* (11) (the optimizer’s coordinates are the very quantities the guarantees are about, so tier 2 holds at every iterate and was adopted by a pre-registered benchmark, not by taste), the *certificate-gated acceptance rule* of section 4.2 (the cheap screens fence, the certificate decides — an uncertified slice cannot become a mark), the *exact raw $\leftrightarrow$ JW conversion with a proved regular domain and a guarded production inverse* (proposition 1; not yet a wired trader workflow — see section 1), and the *analytic LM Jacobian on both charts* that makes SVI the speed peer of the other models.

Limitations, stated plainly. Raw SVI total variance is *one globally convex hyperbola with a single minimum*, so it cannot reproduce a WW-shaped total-variance smile (Note 03) and can be too

rigid for event-shaped targets (Note 01’s double-hat); its single global curve can also over-smooth a sharp short-dated skew. It is butterfly-clean only over a cone the coded fences do not fully certify (section 4), and carries no calendar coupling on its own — that is supplied, softly, by the model-agnostic hinge of Note 10. On sparse chains two further cautions apply: the initializer’s “wing slopes” are finite-span proxies read off the quoted range, not asymptotic observations, and a narrow or one-sided quote set can leave the raw parameters poorly identified — distinct $(a, b, \rho, m, \sigma_{SVI})$ pricing the quoted range indistinguishably — even when the fitted *curve* is stable. Compare curves and handles across fits, not raw coefficients.

8 Traceability

A word on what these tests do and do not prove, so the anchors are not over-read. The benchmark-fit tests exercise an *already tier-2-clean* target, so the “enforcement” rows lock that clean fits are untouched and that the penalty *rows and Jacobians* are correct when activated — they do not demonstrate that an adversarial fit ends feasible. The recovery test whose name says “machine precision” asserts parameter agreement at $\sim 10^{-4}$ relative and curve agreement at $\sim 10^{-6}$ — the freshly generated example itself does reach machine precision, but the *lock* is looser than its name. The conversion-domain contract, previously untested, is now locked by `test_svi_domain.py`, including the classical screens-pass-but- $g < 0$ counterexample.

Table 3: Claims in this note and the code/tests that lock them.

Claim	Object	Code anchor <i>Test anchors</i>
JW→raw conversion exact on the benchmark	section 3	<code>models/svi_jw/svi.py::jw_to_raw</code> <code>test_lqd_pricing.py::test_jw_conversion_matches_note</code>
Regular inverse domain; $\psi = 0$ non-identification; invalid-input behaviour	proposition 1, remark 1	<code>models/svi_jw/svi.py</code> <code>test_svi_domain.py::test_regular_domain_round_trips</code> <code>::test_psi_zero_stratum_not_identified</code> <code>::test_invalid_inputs_fail_loudly_not_plausibly</code>
Core screens do not certify butterfly freedom (adversarial slice)	section 4	<code>models/svi_jw/calibrate.py::_penalties</code> <code>test_svi_domain.py::test_core_screens_do_not_certify_butterfly_freedom</code>
Noise-free benchmark recovery (params $\sim 10^{-4}$ rel, curve $\sim 10^{-6}$)	section 6	<code>models/svi_jw/calibrate.py</code> <code>test_svi_calibrate.py::test_recovers_benchmark_parameters</code> <code>::test_curve_reproduced_to_machine_precision</code> (name predates the looser lock)
Clean fits untouched by the fences (benchmark is tier-2-clean)	proposition 2	<code>models/svi_jw/calibrate.py::_penalties</code> <code>test_svi_calibrate.py::test_respects_lee_wing_bound</code> <code>::test_min_variance_non_negative</code>

Claim	Object	Code anchor <i>Test anchors</i>
Analytic Jacobian matches FD, section 5.4 penalty rows active or not (row correctness, not end-state feasibility)		models/svi_jw/jacobian.py test_svi_jacobian.py::test_mid_fit_admissible ::test_lee_penalty_active ::test_min_variance_penalty_active
Calendar hinge rows correct; off is byte-identical	section 5	models/svi_jw/calibrate.py test_svi_jacobian.py::test_calendar_floor_active test_overlay_calendar.py::test_svi_no_floor_is_byte_identical
Arbitrary per-quote weights accepted (low-level API; not a surfaced setting)	section 5	models/svi_jw/calibrate.py::vega_weights test_svi_calibrate.py::test_recovers_under_vega_weights
Prior blocks reach the SVI overlay	section 5	models/svi_jw/calibrate.py test_prior_parametric.py::test_operator_prior_pulls_all_models_toward_prior_skew
Structural chart: ratified default; round trip exact; fenced without penalties; charts agree on clean quotes	section 5.2	models/svi_jw/structural.py test_svi_structural_chart.py::test_structural_is_the_ratified_default ::test_chart_round_trip_and_identities ::test_structural_chart_is_fenced_without_penalties ::test_charts_agree_on_clean_quotes
Belly certificate: Vogt slice fails where screens pass; uncertified blocks publish; repair certifies	section 4.2	models/diagnostics.py::belly_certificate test_belly_certificate.py::test_vogt_slice_fails_certificate_where_screens_pass ::test_uncertified_belly_blocks_publish test_svi_structural_chart.py::test_vogt_quotes_are_repaired_and_certified
Lee cap strictly buffered; $\beta = 2$ boundary admits negative tail density	section 4.4	models/svi_jw/calibrate.py test_svi_lee_boundary.py::test_beta2_boundary_admits_negative_tail_density ::test_default_cap_is_strictly_buffered ::test_fit_toward_the_boundary_is_fenced_at_the_buffer
Adversarial battery: finite or refuses with a reason, both charts	section 5	models/svi_jw/calibrate.py test_svi_adversarial.py::test_three_quotes_fits_and_two_refuses ::test_evaluation_cap_exhaustion_is_honest

Claim	Object	Code anchor <i>Test anchors</i>
Guarded converter: structured failures, stable near the stratum; desk ticket and forward bump	section 3.3, section 3.4	models/svi_jw/svi.py::jw_to_raw_checked test_svi_desk_and_guards.py::test_guarded_converter_structured_failures ::test_guarded_converter_is_stable_near_the_stratum ::test_desk_ticket_reads_the_instruments ::test_forward_bump_moves_desk_quantities_on_a_skewed_smile

A Hyperparameter atlas

Table 4: SVI hyperparameters.

Knob	Default	Role
<i>Surfaced (FitSettings / OptionsSettings)</i>		
sviChart	structural	Optimization chart (section 5.2): every iterate strictly floor-positive and strictly Lee-clean, penalties inert; raw = the historical vector, kept for comparability and rollback.
bellyRepair	on	One-shot belly-hinge repair refit when the certificate fails; the repair is kept only if it certifies (section 4.2).
sviPenaltyWeight W	1000	Residual multiplier of the two soft no-arbitrage rows (9) (effective W^2 in the squared cost; 0 = fences off).
leeSlopeMax $\beta_{\max}$	1.95	Lee wing-slope cap, strictly buffered under Lee’s bound — $\beta = 2$ itself admits negative tail density (section 4.4); values ≥ 2 are accepted as explicit configuration and restore the broken boundary.
midAnchorWeight weightScheme	0.05 equal	Mid anchor weight in the band-fit objective (Note 07). Per-quote weights, equal / tv_density (Note 07). Arbitrary weights (e.g. vega ²) exist only at the low-level API, not as a setting.
calendarWeight	10^6	Calendar hinge penalty (Note 10; lives in OptionsSettings , not FitSettings).
extrapEnforce	off	Tapered extrapolated-region fences (section 4.3; Notes 09–10). Calibration-affecting; hybrid Jacobian when on.
<i>Internal (reparametrization / solver)</i>		
$b = \text{softplus}(\theta_b)$	—	Enforces $b \geq 0$.
$\rho = \tanh(\theta_\rho)$	—	Enforces $ \rho < 1$.
$\sigma_{\text{SVI}} = e^{\theta_\sigma}$	—	Enforces $\sigma_{\text{SVI}} > 0$.

Knob	Default	Role
LM tolerances	10^{-15}	<code>xtol/ftol/gtol</code> ; LM converges in few iterations so tight tolerances are cheap.
floor 10^{-12}	—	Total-variance floor inside $\sqrt{w/t}$ to avoid $\sqrt{\text{neg}}$ on structurally-valid-but-negative-variance trial slices.

B Performance notes

1. **Levenberg–Marquardt over trust-region.** On real noisy chains LM crosses the penalty kinks in fewer iterations than `trf` at matched tolerance; `trf` at 10^{-10} was measured *slower* on real nodes.
2. **Analytic Jacobian.** Closed-form Jacobian of the core residual stack (data + penalties + calendar) through the reparametrization. Two measurements, with their scopes: the note’s *residual/Jacobian microbenchmark* (25 synthetic quotes, core rows only, single-threaded development machine, fastest of three) measured 0.9 ms vs 2.5 ms ($2.85\times$) with the costs agreeing to $2.7e-33$ — same solution within fit precision, not bit-identical; the *historical real-node* measurement (spike-regime backtest nodes, June-2026 harness) was $26.3 \rightarrow 10.2$ ms per fit ($\approx 2.6\times$) — the finding that motivated shipping it. Gated to the var-swap/prior-free configuration; hybrid (analytic core + FD extrapolation rows) under `extrapEnforce`.
3. **Structural validity by construction.** The reparametrization removes the box-constraint handling entirely, so each LM step is a plain linear solve.
4. **Geometric warm start.** Reading (a, m) from the vertex and (b, ρ) from the wing slopes lands θ_0 close enough that liquid smiles converge in a single pass.
5. **Structural-chain Jacobian** (adoption follow-up, 2026-07-26). The structural chart’s rows are assembled in raw space and pushed through the closed-form 5×5 chain matrix; measured on the adoption A/B: identical smiles at $2.1\text{--}2.4\times$ over the FD structural path (lab $4.74 \rightarrow 1.98$ ms, boundary-pull $2.67 \rightarrow 1.29$ ms, Vogt $1.73 \rightarrow 0.81$ ms), $\text{nfev} \leq 26$ vs 30–86. Even before this, the structural chart was $\sim 3\times$ faster than raw on real nodes — raw burns its evaluation budget on a third of real fits (section 5.2).

C Reference implementation: reverse map and Jacobian core

The raw slice evaluates in one line, $w(k) = a + b(\rho(k - m) + \sqrt{(k - m)^2 + \sigma_{\text{SVI}}^2})$; the reverse conversion (4) reads the five jump-wing handles off it. Composing with listing 1 recovers the original handles to 10^{-10} .

Listing 2: The reverse map raw $\rightarrow$ SVI-JW — equations (4). The backend ships only the forward converter; this reverse listing is the one the figure generator runs.

```

1 def w_svi(k, a, b, rho, m, sig):
2     # the hyperbola w(k); sig is the SVI width
3     km = k - m
4     return a + b*(rho*km + np.sqrt(km*km + sig**2))

```

```

5
6 def raw_to_jw(t, a, b, rho, m, sig):
7     # (a, b, rho, m, sig) -> (v, psi, p, c, vtilde)
8     w0 = w_svi(0.0, a, b, rho, m, sig); sq = np.sqrt(w0)
9     v = w0/t
10    psi = b/(2*sq)*(rho - m/np.sqrt(m*m + sig**2))
11    p, c = b*(1 - rho)/sq, b*(1 + rho)/sq
12    vt = (a + b*sig*np.sqrt(1 - rho*rho))/t
13    return v, psi, p, c, vt

```

The production Jacobian core (section 5.4) is equally short — the raw partials (13) times the reparametrization factors (15), then the IV chain rule (14) for the data rows (distilled from `models/svi_jw/jacobian.py`, which it matches to machine precision):

Listing 3: The analytic Jacobian’s core: $dw/d\theta$ after the chain rule, and the IV data rows.

```

1 def dw_dtheta(k, a, b, rho, m, sig):
2     # d w / d theta, theta = (a, th_b, th_rho, m, th_sig)
3     x = k - m
4     r = np.sqrt(x*x + sig*sig)
5     sig_b = 1.0 - np.exp(-b) # d softplus / d th_b
6     J = np.empty((k.size, 5))
7     J[:, 0] = 1.0 # d/da
8     J[:, 1] = (rho*x + r)*sig_b # d/db . db/dth_b
9     J[:, 2] = (b*x)*(1 - rho*rho) # d/drho . drho/dth_rho
10    J[:, 3] = -b*(rho + x/r) # d/dm
11    J[:, 4] = (b*sig/r)*sig # d/dsig . dsig/dth_sig
12    return J
13
14 def data_rows(k, t, w, J, sqrt_weights):
15     # residual = sqrt(w_i) (sqrt(w(k)/t) - vol_i): chain rule
16     model_vol = np.sqrt(np.maximum(w, 1e-12)/t)
17     rows = J / (2.0*t*model_vol)[: , None]
18     rows[w <= 1e-12] = 0.0 # the floor flattens the gradient
19     return sqrt_weights[: , None] * rows

```

References

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